首都经济贸易大学
CAPITAL UNIVERSITY OF ECONOMICS AND BUSINESS

Unified Training of Universal Time Series Forecasting Transformers

通用时间序列预测变压器的统一训练

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Abstract

Deep learning for time series forecasting has traditionally operated within a one-model-per-dataset framework, limiting its potential to leverage the game-changing impact of large pre-trained models. The concept of *universal forecasting*, emerging from pre-training on a vast collection of time series datasets, envisions a single Large Time Series Model capable of addressing diverse downstream forecasting tasks. However, constructing such a model poses unique challenges specific to time series data: i) cross-frequency learning, ii) accommodating an arbitrary number of variates for multivariate time series, and iii) addressing the varying distributional properties inherent in large-scale data. To address these challenges, we present novel enhancements to the conventional time series Transformer architecture, resulting in our proposed **M**asked Enc**O**der-based **U**n**I**ve**R**s**A**l **T**Ime Series Forecasting Transformer (**MOIRAI**). Trained on our newly introduced Large-scale Open Time Series Archive (LOTSA) featuring over 27B observations across nine domains, MOIRAI achieves competitive or superior performance as a zero-shot forecaster when compared to full-shot models. Code, data, and model weights can be found at https://github.com/SalesforceAIResearch/uni2ts.

通用预测范式universal forecasting

- 传统时序预测——一对一
- 通用时序预测

（1）现阶段存在的问题：

1）交叉频率学习

2）适应多变量时间序列的任意数量的变量

3）处理大规模数据中固有的**不同分布**特性

- 作者的工作

——提出Moirai模型（基于Transformer架构改编的掩码编码器

——引入大规模开源时序档案（LOTSA）

- 预训练代码 https://github.com/SalesforceAIResearch/uni2ts
- 预训练数据集 *https://huggingface.co/datasets/Salesforce/lotsa_data*

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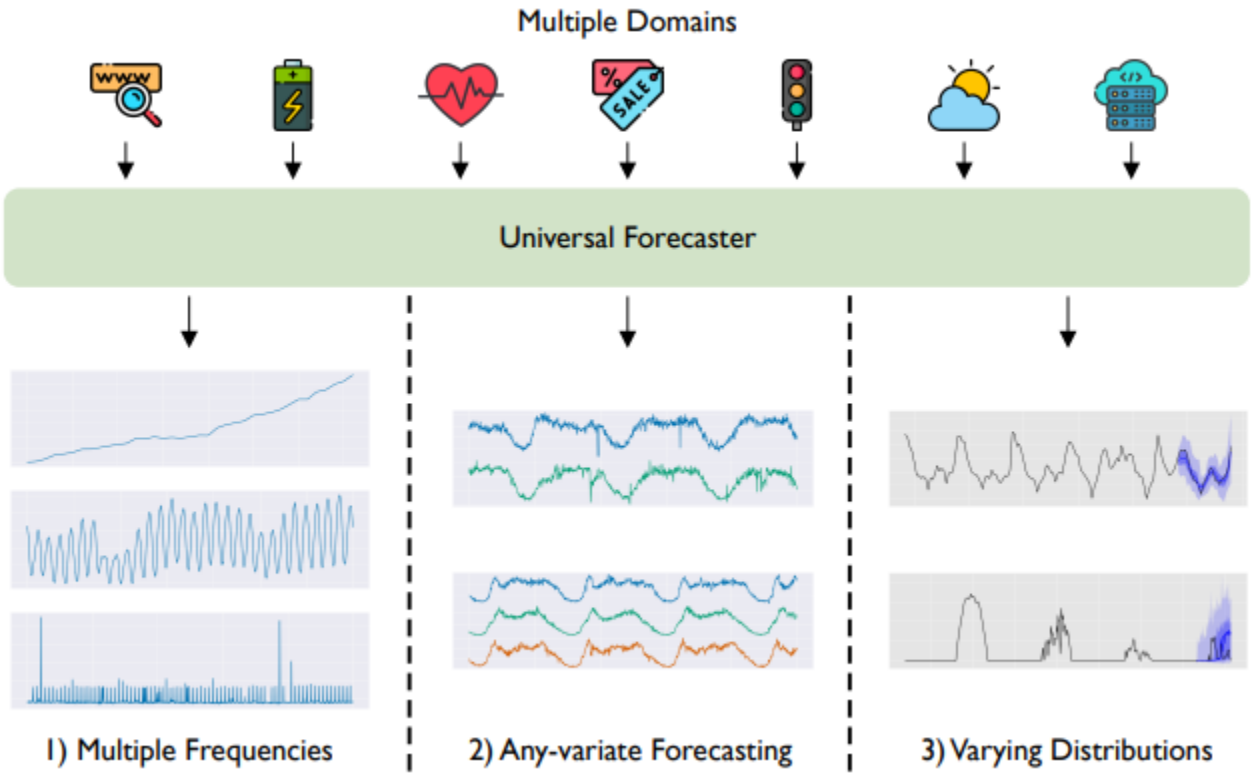
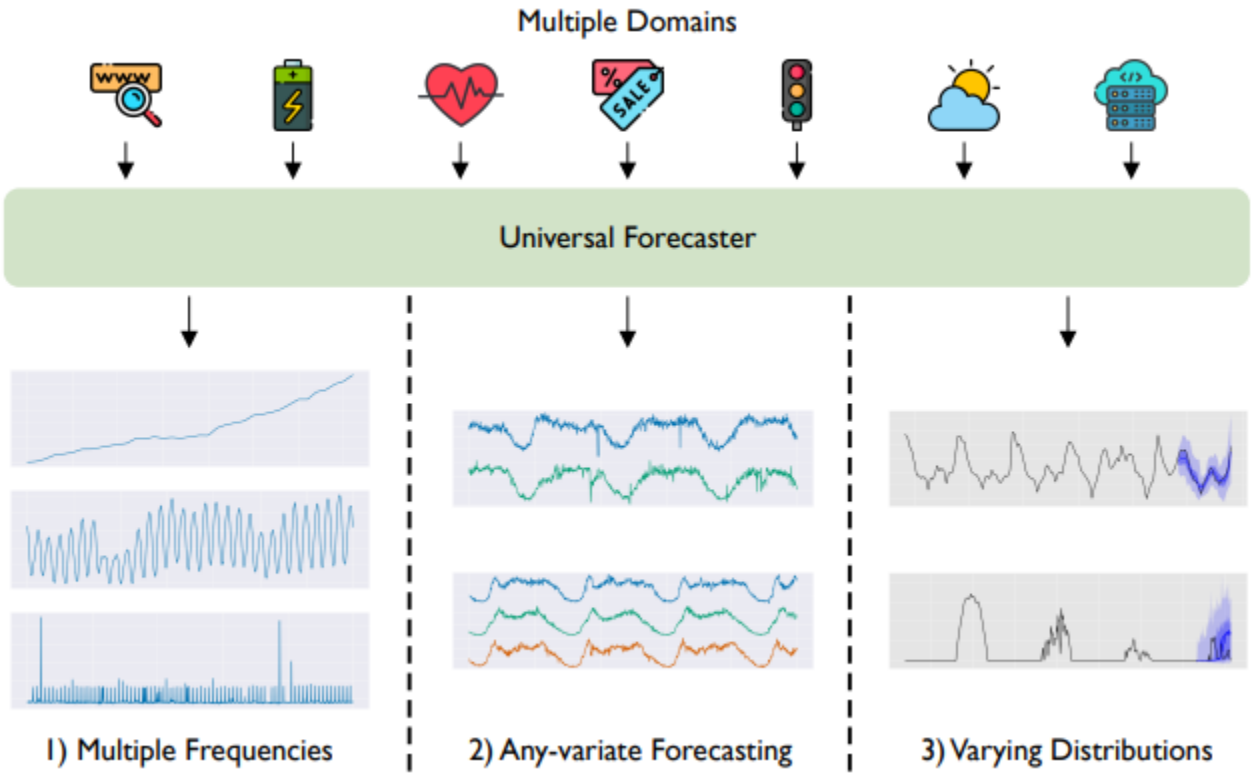
01 Introduction



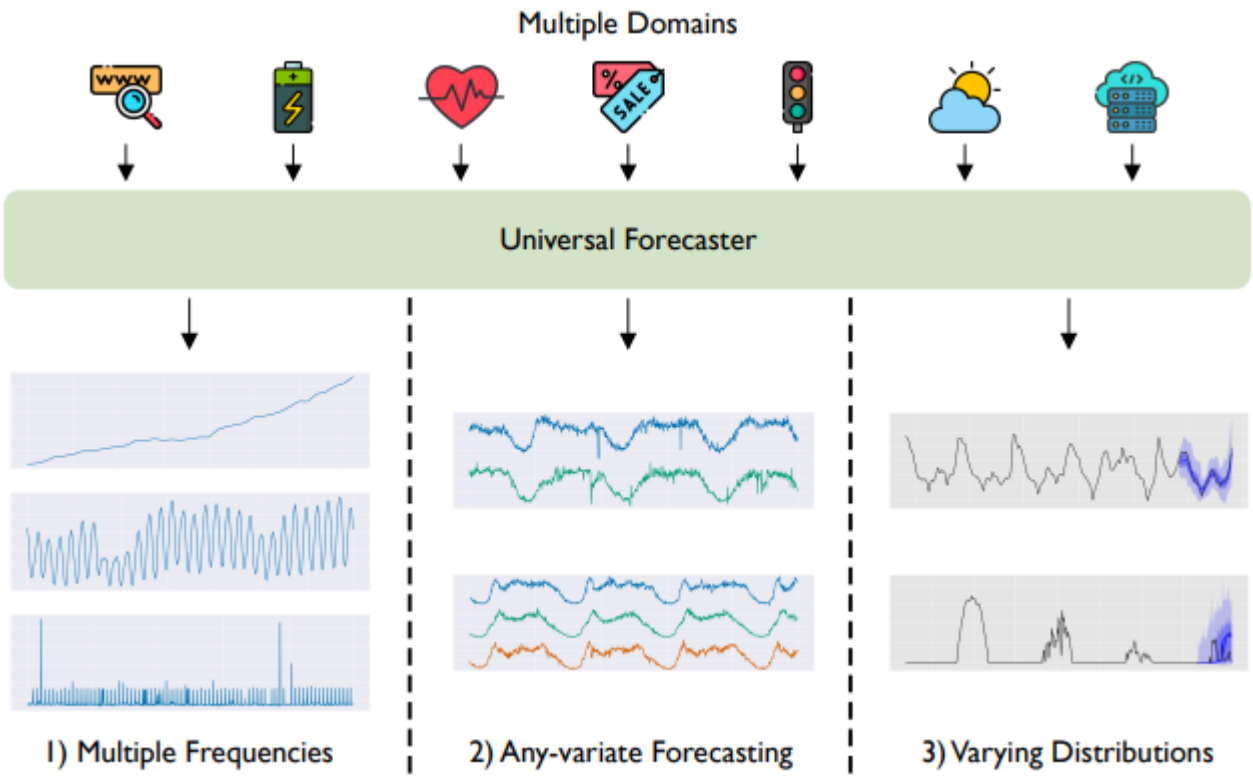
时间序列的频率在确定其存在的模式方面起着重要作用。
由于时序负干扰，多频率学习十分具有挑战性，现有工作通常为每个频率训练一个模型来解决问题。

01 Introduction



Multi Patch Size Output/Input Projections
学习多个输入输出投影层
learn multiple input and output projection layers

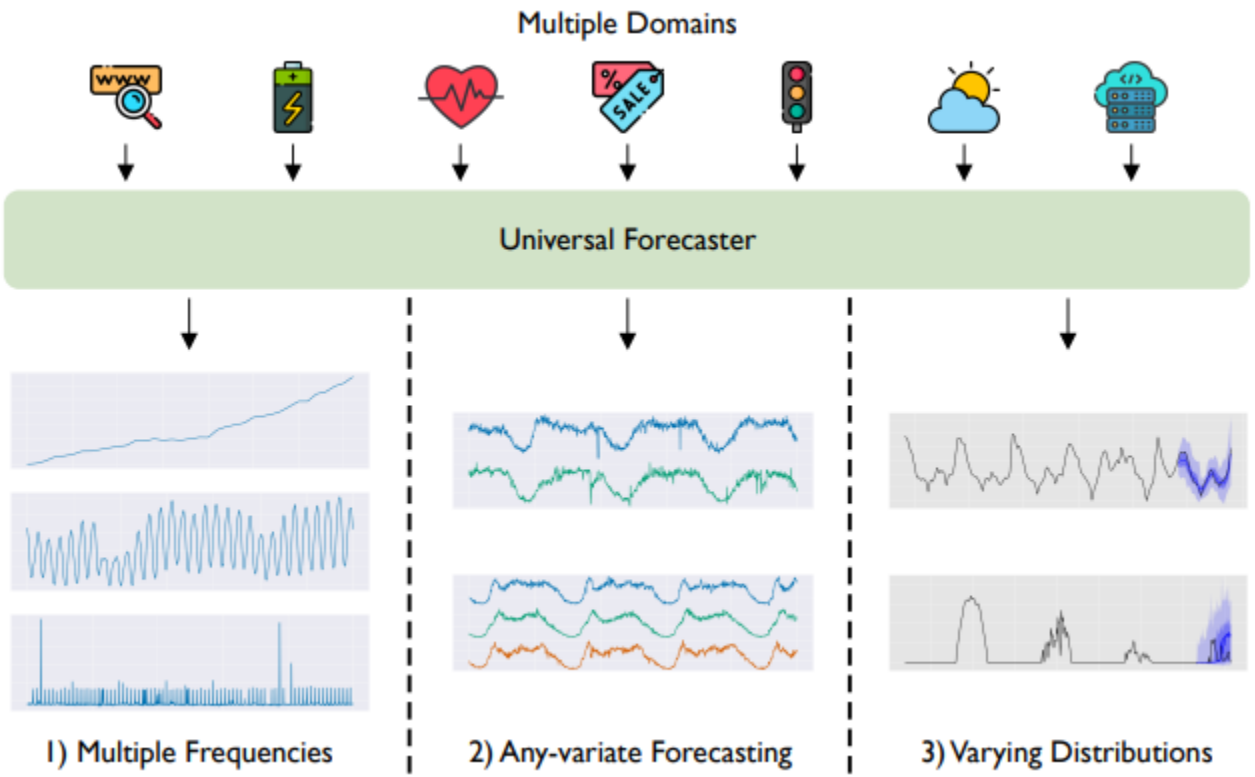
01



时间序列数据在维度上是异质的，其中多元时间序列可以有不同数量的变量
每个变量在跨数据集时测量的是语义上不同的量

Any-variate Attention 任意变量注意力机制

01



大规模数据中的不同分布特性
——例如，使用对称分布（如正态分布、t分布）作为预测分布不适合正时间序列——使得预先定义简单参数分布的标准方法不足以灵活地捕获各种数据集。

MIXTURE DISTRIBUTION
混合分布

02 Related Work

Table 1. Comparison between pre-trained forecasting models. Further discussion on the notion of a "flexible distribution" can be found in Appendix B.3.

	Any-variate (Zero-shot)	Probabilistic Forecasting	Flexible Distribution	Pre-training Data (Size)	Open-source
MOIRAI	✓	✓	✓	LOTSA (> 27B)	✓
TimeGPT-1	✓	✓	✗	Unknown (100B)	✗
ForecastPFN	✗	✗	-	Synthetic Data (60M)	✓
Lag-Llama	✗	✓	✗	Monash (< 1B)	✓
TimesFM	✗	✗	-	Wiki + Trends + Others (> 100B)	✓
TTM	✗	✗	-	Monash (< 1B)	✓
LLMTime	✗	✓	✓	Web-scale Text	✓

zero-shot forecasting capabilities

Pre-training for Zero-shot Forecasting

03 Method

Problem Formulation Consider a dataset of N time series $\mathcal{D} = \{(\boldsymbol{Y}^{(i)}, \boldsymbol{Z}^{(i)})\}_{i=1}^{N}$, where $\boldsymbol{Y}^{(i)} = (\boldsymbol{y}_1^{(i)}, \boldsymbol{y}_2^{(i)}, \ldots, \boldsymbol{y}_{T_i}^{(i)}) \in \mathbb{R}^{d_{y_i} \times T_i}$ is a target time series of d_{y_i} variates and T_i time steps. Each time series is associated with a set of covariates $\boldsymbol{Z}^{(i)} = (\boldsymbol{z}_1^{(i)}, \boldsymbol{z}_2^{(i)}, \ldots, \boldsymbol{z}_{T_i}^{(i)}) \in \mathbb{R}^{d_{z_i} \times T_i}$. The goal is to forecast the predictive distribution $p(\boldsymbol{Y}_{t:t+h}|\boldsymbol{\phi})$ by predicting distribution parameters $\boldsymbol{\phi}$ via a learned model $f_{\boldsymbol{\theta}} : (\boldsymbol{Y}_{t-l:t}, \boldsymbol{Z}_{t-l:t+h}) \mapsto \hat{\boldsymbol{\phi}}$ which maximizes the log-likelihood:

$$\max_{\boldsymbol{\theta}} \mathop{\mathbb{E}}_{\substack{(\mathbf{Y},\mathbf{Z})\sim p(\mathcal{D}) \\ (\mathrm{t,l,h})\sim p(\mathcal{T}|\mathcal{D})}} \log p(\mathbf{Y}_{\mathrm{t:t+h}}|\hat{\boldsymbol{\phi}}),$$

$$\text{s.t. } \hat{\boldsymbol{\phi}} = f_{\boldsymbol{\theta}}(\mathbf{Y}_{\mathrm{t-l:t}}, \mathbf{Z}_{\mathrm{t-l:t+h}}), \tag{1}$$

where $p(\mathcal{D})$ is the data distribution which samples for a time series, $(\boldsymbol{Y}, \boldsymbol{Z})$, and $p(\mathcal{T}|\mathcal{D})$ is the task distribution which defines the lookback window, $\boldsymbol{Y}_{t-l:t} = (\boldsymbol{y}_{t-l}, \ldots, \boldsymbol{y}_{t-1})$ with context length l and forecast horizon, $\boldsymbol{Y}_{t:t+h} = (\boldsymbol{y}_t, \ldots, \boldsymbol{y}_{t+h-1})$ with prediction length h.

03 Method

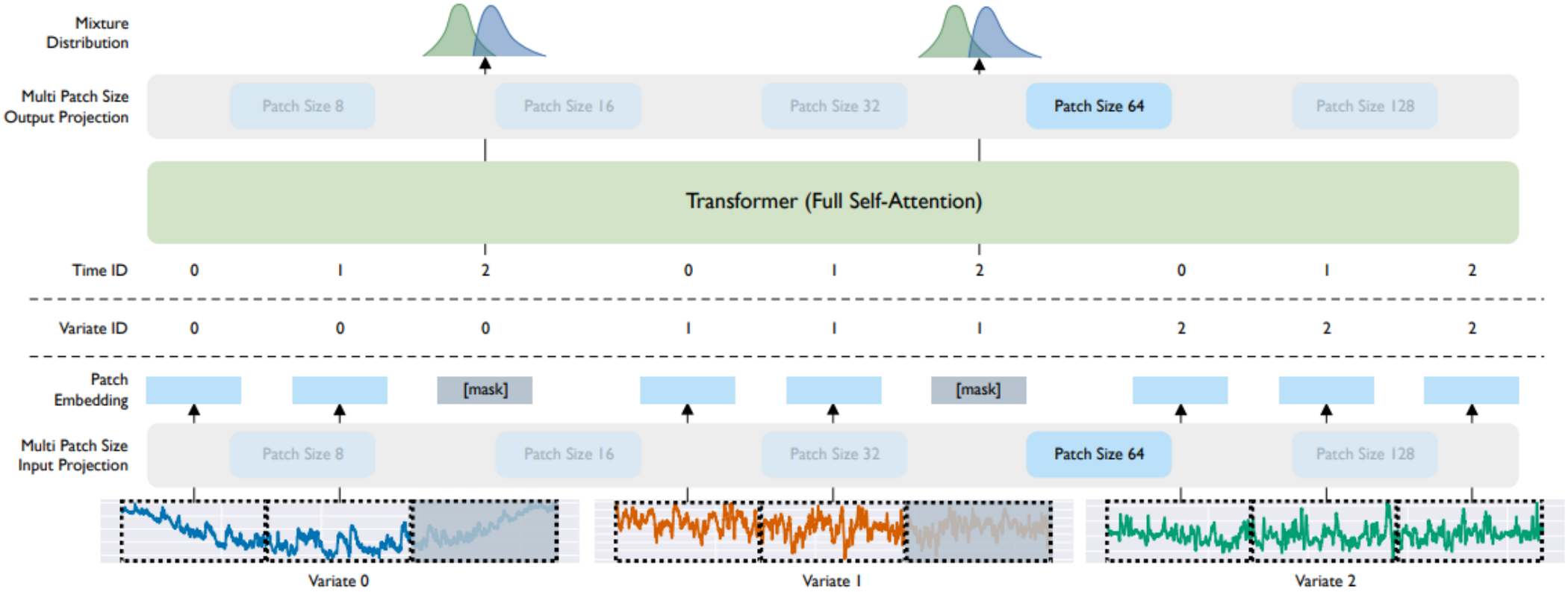


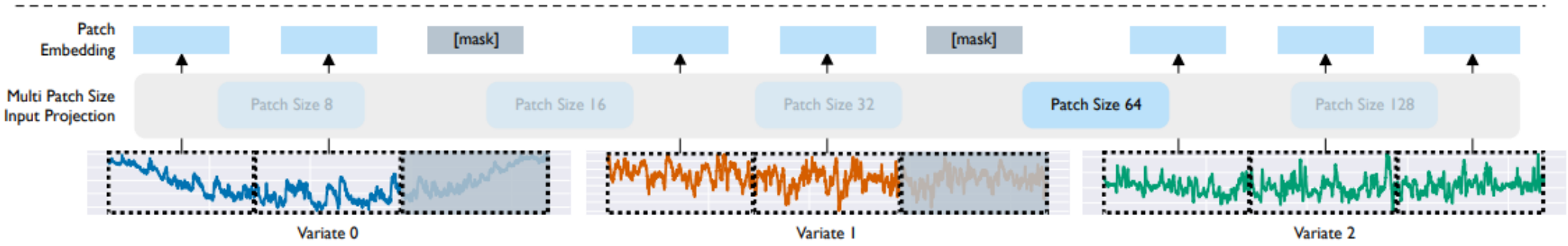
Figure 2. Overall architecture of MOIRAI. Visualized is a 3-variate time series, where variates 0 and 1 are target variables (i.e. to be forecasted, and variate 2 is a dynamic covariate (values in forecast horizon known). Based on a patch size of 64, each variate is patchified into 3 tokens. The patch embeddings along with sequence and variate id are fed into the Transformer. The shaded patches represent the forecast horizon to be forecasted, whose corresponding output representations are mapped into the mixture distribution parameters.

在input/output还做了instance normalization

03 Multi Patch Size Output/Input Projections

from the prevailing one-model-per-dataset paradigm. Instead, we aim for a more flexible strategy: opting for a larger patch size to handle high-frequency data, thereby lower the burden of the quadratic computation cost of attention while maintaining a long context length. Simultaneously, we advocate for a smaller patch size for low-frequency data to transfer computation to the Transformer layers, rather than relying solely on simple linear embedding layers. To implement this approach, we propose learning multiple input and output embedding layers, each associated with varying patch sizes. The selection of the appropriate patch size for a given time series frequency relies on pre-defined settings (see Appendix B.1). Note that we only learn one set of projection weights per patch size, which is shared amongst frequencies if there is an overlap based on the settings.

- Yearly, Quarterly: 8
- Monthly: 8, 16, 32
- Weekly, Daily: 16, 32
- Hourly: 32, 64
- Minute-level: 32, 64, 128
- Second-level: 64, 128



03 ANY-VARIATE ATTENTION

将时间轴和变量轴视为单个序列，并利用 RoPE（Rotary Position Embedding，旋转位置嵌入）和可学习的二元注意力偏差来分别对时间轴和变量轴进行编码

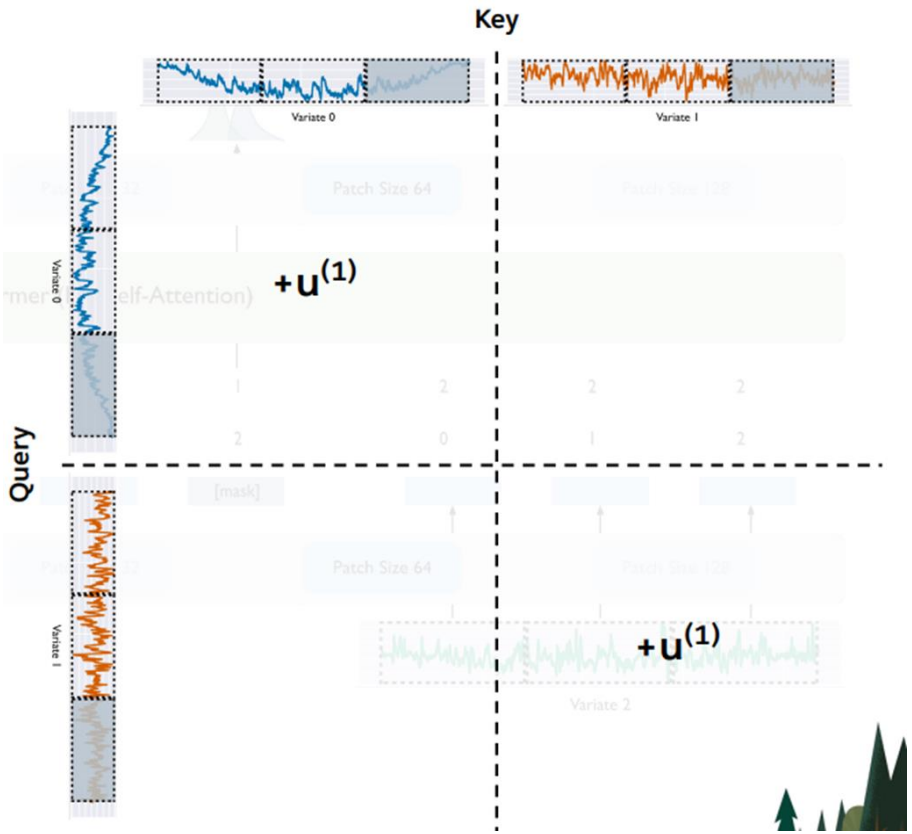
- Attention score between (i,m)-th query and (j,n)-th key
  - i,j: time index
  - n,m: variate index

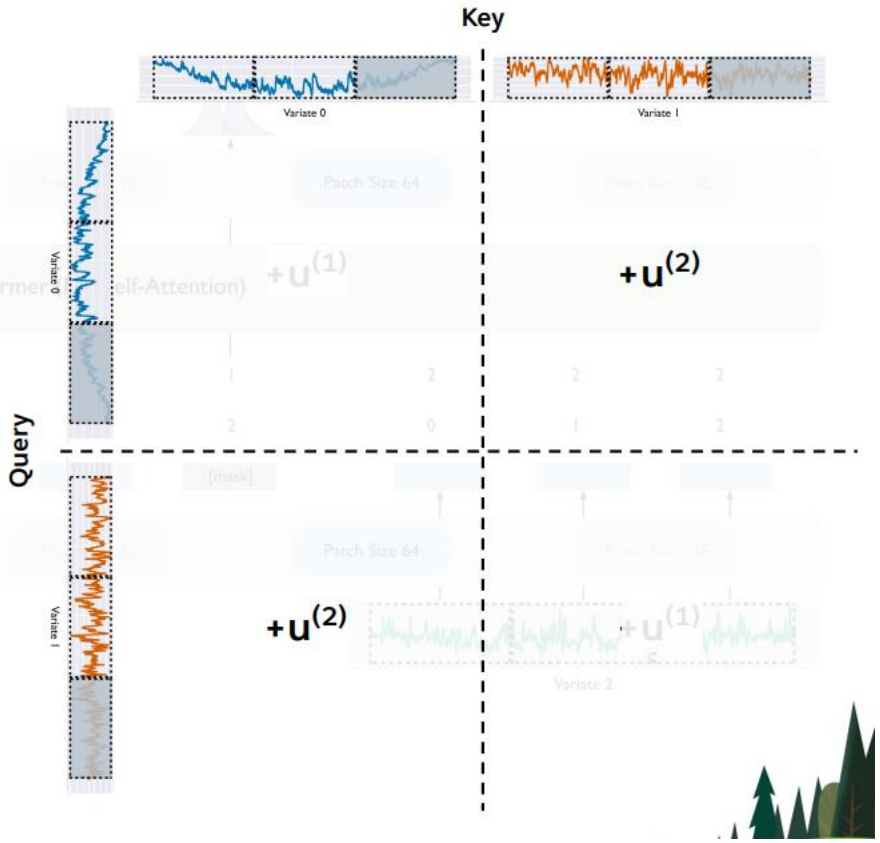
$$E_{ij,mn} = \overbrace{(\boldsymbol{W}^{Q}\boldsymbol{x}_{i,m})^{T}\boldsymbol{R}_{i-j}(\boldsymbol{W}^{K}\boldsymbol{x}_{j,n})}^{\text{Rotary matrix (RoPE)}} \underbrace{+ u^{(1)} * \mathbb{1}_{\{m=n\}} + u^{(2)} * \mathbb{1}_{\{m\neq n\}}}_{\text{Binary attention bias}},$$

二元注意偏差分量通过注意分数来消除变量之间的歧义，并且可以扩展到任意数量的变量。

$$A_{ij,mn} = \frac{\exp\{E_{ij,mn}\}}{\sum_{k,o}\exp\{E_{ik,mo}\}},$$

03 ANY-VARIATE ATTENTION





$$\mathbb{1}_{\{\text{cond}\}} = \begin{cases} 1, \text{ if cond} \\ 0, \text{ otherwise} \end{cases}$$

03 MIXTURE DISTRIBUTION

$$p(\mathbf{Y}_{t:t+h}|\hat{\boldsymbol{\phi}}) = \sum_{i=1}^{c} w_i p_i(\mathbf{Y}_{t:t+h}|\hat{\boldsymbol{\phi}}_i)$$

From the output projection layer

where $\hat{\boldsymbol{\phi}} = \{w_1, \hat{\boldsymbol{\phi}}_1, \ldots, w_c, \hat{\boldsymbol{\phi}}_c\}$, and p_i is the i-th component's p.d.f. While the choice of mixture components

03 Transformer

RMSNorm替换所有layernorm， 并应用查询键归一化。用SwiGLU 代替FFN层（前馈神经网络层）中的非线性，调整隐藏维度， 使其具有与原始FFN层相同数量的参数。

在 Transformer模块的所有层中都忽略了偏差。

03 Transformer模块

3.1. Architecture

Illustrated in Figure 2, MOIRAI follows a (non-overlapping) patch-based approach to modeling time series with a masked encoder architecture. One of our proposed modifications to extend the architecture to the any-variate setting is to "flatten" multivariate time series, considering all variates as a single sequence. Patches are subsequently projected into vector representations via a multi patch size input projection layer. The [mask] signifies a learnable embedding which replaces patches falling within the forecast horizon. The output tokens are then decoded via the multi patch size output projection into the parameters of the mixture distribution. While not visualized, (non-learnable) instance normalization (Kim et al., 2022) is applied to inputs/outputs, aligning with the current standard practice for deep forecasting models.

The core Transformer module is an encoder-only Transformer architecture, leveraging various improvements as proposed by recent state-of-the-art LLM architectures. We use pre-normalization (Xiong et al., 2020) and replace all LayerNorms with RMSNorm (Zhang & Sennrich, 2019), and also apply query-key normalization (Henry et al., 2020). The non-linearity in FFN layers are replaced with SwiGLU (Shazeer, 2020), adjusting the hidden dimension to have equal number of parameters as the original FFN layer. We omit biases in all layers of the Transformer module.

MOIRAI遵循一种(非重叠)基于补丁的方法，
使用掩码编码器架构对时间序列进行建模。我们提出的将该架构扩展到任意变量设置的修改之一是“平坦化”多元时间序列，将所有变量视为单个序列。随后，通过多patch大小的输入投影层，将patch投影成向量表示。[mask]表示一个可学习的嵌入，它取代落在预测范围内的补丁。
通过多patch大小的输出投影将输出 令牌解码为混合分布的参数。

03 LOTSA

Table 2. Key statistics of LOTSA by domain.

	Energy	Transport	Climate	CloudOps	Web	Sales	Nature	Econ/Fin	Healthcare
# Datasets	30	23	6	3	3	6	5	23	6
# Obs.	16,358,600,896	4,900,453,419	4,188,011,890	1,518,268,292	428,082,373	197,984,339	28,547,647	24,919,596	1,594,281
%	59.17%	17.73%	15.15%	5.49%	1.55%	0.72%	0.09%	0.10%	0.01%

Table 3. Key statistics of LOTSA by frequency.

	Yearly	Quarterly	Monthly	Weekly	Daily	(Multi) Hourly	(Multi) Minute-level	(Multi) Second-level
# Datasets	4	5	10	7	21	31	25	2
# Obs.	873,297	2,312,027	11,040,648	18,481,871	709,017,118	19,875,993,973	7,013,949,430	14,794,369
%	0.003%	0.008%	0.040%	0.067%	2.565%	71.893%	25.370%	0.054%

在它们上训练LTM可能会导致误导性的结论

the Monash Time Series Forecasting Archive (Godahewa et al., 2021), datasets provided by the GluonTS library (Alexandrov et al., 2020), and datasets from the popular long sequence forecasting benchmark (Lai et al., 2018; Wu et al., 2021).

03 Pre-training

Table 4. Details of MOIRAI model sizes.

	Layers	d_{model}	d_{ff}	Heads	d_{kv}	Params
MOIRAI$_{\text{Small}}$	6	384	1536	6	64	14m
MOIRAI$_{\text{Base}}$	12	768	3072	12	64	91m
MOIRAI$_{\text{Large}}$	24	1024	4096	16	64	311m

03 Pre-training

- 数据分布〇由于数据不平衡导致的超大数据集的上限采样百分比
- 任务分配〇随机样本上下文长度，预测长度〇随机子样本多元时间序列〇将对齐的单变量时间序列随机组合为多变量时间序列

03 Pre-training

1. 数据分布设计

通过将数据分布分解为子数据集分布和基于子数据集的时间序列分布，进而引入了子数据集的概念。

研究者首先从 p(D) 中抽取一个子数据集，然后基于该子数据集抽取一个时间序列。然而，由于不同领域和频率之间的数据不平衡，为避免按比例抽取子数据集，而是首先限制每个子数据集的贡献在 $\epsilon = 0.001$ 之内，然后再进行归一化。

2. 任务分布设计

目标是训练一个能够在不同上下文和预测长度下具有预测能力的模型。
因此，不固定上下文和预测长度，从任务分布中采样，该分布定义了给定时间序列的回溯窗口和预测范围。

每个变量 的最小序列长度为2，总最大序列长度为512

04 In-distribution Forecasting

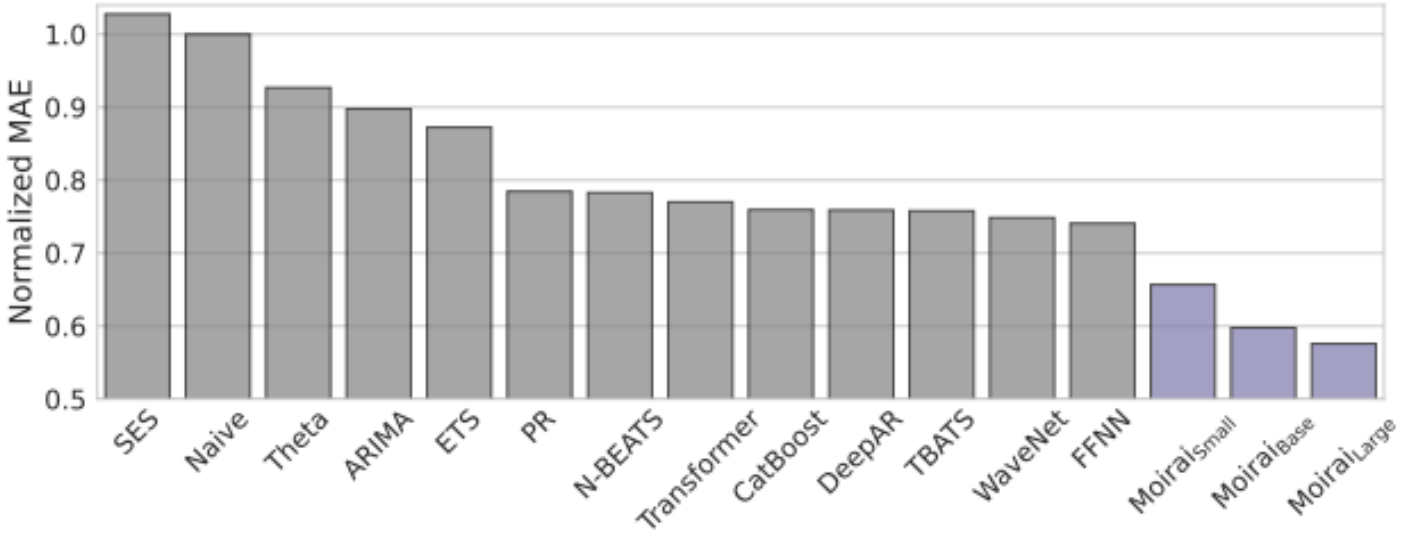


Figure 3. Aggregate results of the Monash Time Series Forecasting Benchmark. The normalized MAE is reported, which normalizes

归一化平均绝对误差(MAE)

在通过 Monash 时序预测基准上进行 in-distribution 评估时，MOIRAI 模型击败了所有的基准模型。

04 Out of distribution Forecasting

Table 5. Probabilistic forecasting results. Best results are highlighted in **bold**, and second best results are underlined. Baseline results are aggregated over five training runs with different seeds, reporting the mean and standard deviation.

		Zero-shot			Full-shot				Baseline	
		MOIRAI$_{\text{Small}}$	MOIRAI$_{\text{Base}}$	MOIRAI$_{\text{Large}}$	PatchTST	TiDE	TFT	DeepAR	AutoARIMA	Seasonal Naive
Electricity	CRPS	0.072	0.055	0.050	0.052±0.00	**0.048±0.00**	0.050±0.00	0.065±0.01	0.327	0.070
	MSIS	7.999	6.172	5.875	5.744±0.12	**5.672±0.08**	6.278±0.24	6.893±0.82	29.412	35.251
Solar	CRPS	0.471	0.419	**0.406**	0.518±0.09	0.420±0.00	0.446±0.03	0.431±0.01	1.055	0.512
	MSIS	8.425	7.011	**6.250**	8.447±1.59	13.754±0.32	8.057±3.51	11.181±0.67	25.849	48.130
Walmart	CRPS	0.103	0.093	0.098	0.082±0.01	**0.077±0.00**	0.087±0.00	0.121±0.00	0.124	0.151
	MSIS	9.371	8.421	8.520	**6.005±0.21**	6.258±0.12	8.718±0.10	12.502±0.03	9.888	49.458
Weather	CRPS	0.049	**0.041**	0.051	0.059±0.01	0.054±0.00	0.043±0.00	0.132±0.11	0.252	0.068
	MSIS	5.236	5.136	**4.962**	7.759±0.49	8.095±1.74	7.791±0.44	21.651±17.34	19.805	31.293
Istanbul Traffic	CRPS	0.173	0.116	0.112	0.112±0.00	0.110±0.01	0.110±0.01	**0.108±0.00**	0.589	0.257
	MSIS	5.937	4.461	4.277	**3.813±0.09**	4.752±0.17	4.057±0.44	4.094±0.31	16.317	45.473
Turkey Power	CRPS	0.048	0.040	**0.036**	0.054±0.01	0.046±0.01	0.039±0.00	0.066±0.02	0.116	0.085
	MSIS	7.127	6.766	**6.341**	8.978±0.51	8.579±0.52	7.943±0.31	13.520±1.17	14.863	36.256

04 Long Sequence Forecasting

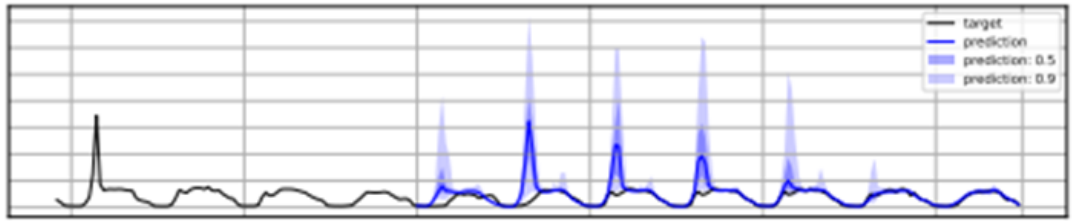
Table 6. Long sequence forecasting results. Results are averaged across prediction lengths {96, 192, 336, 720}. Best results are highlighted in **bold**, and second best results are underlined. Full-shot results are obtained from Liu et al. (2023b).

		Zero-shot			Full-shot							
		MOIRAI$_{Small}$	MOIRAI$_{Base}$	MOIRAI$_{Large}$	iTransformer	TimesNet	PatchTST	Crossformer	TiDE	DLinear	SCINet	FEDformer
ETTh1	MSE	**0.400**	0.434	0.510	0.454	0.458	0.469	0.529	0.541	0.456	0.747	0.44
	MAE	**0.424**	0.438	0.469	0.448	0.450	0.455	0.522	0.507	0.452	0.647	0.46
ETTh2	MSE	**0.341**	0.345	0.354	0.383	0.414	0.387	0.942	0.611	0.559	0.954	0.437
	MAE	0.379	0.382	**0.376**	0.407	0.497	0.407	0.684	0.550	0.515	0.723	0.449
ETTm1	MSE	0.448	**0.381**	0.390	0.407	0.400	0.387	0.513	0.419	0.403	0.486	0.448
	MAE	0.409	**0.388**	0.389	0.410	0.406	0.400	0.495	0.419	0.407	0.481	0.452
ETTm2	MSE	0.300	**0.272**	0.276	0.288	0.291	0.281	0.757	0.358	0.35	0.571	0.305
	MAE	0.341	0.321	**0.320**	0.332	0.333	0.326	0.611	0.404	0.401	0.537	0.349
Electricity	MSE	0.233	0.188	0.188	**0.178**	0.193	0.216	0.244	0.252	0.212	0.268	0.214
	MAE	0.320	0.274	0.273	**0.270**	0.295	0.304	0.334	0.344	0.3	0.365	0.327
Weather	MSE	0.242	**0.238**	0.259	0.258	0.259	0.259	0.259	0.271	0.265	0.292	0.309
	MAE	0.267	**0.261**	0.275	0.278	0.287	0.281	0.315	0.320	0.317	0.363	0.36

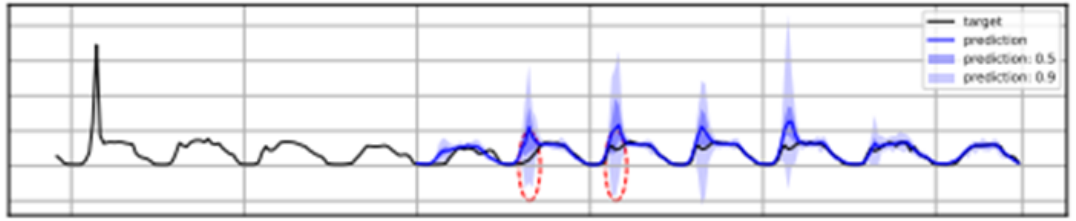
04 消融实验

Table 7. Ablation study on Monash benchmark. The aggregated normalized MAE, similarly calculated as in Figure 3 is reported.

	Normalized MAE
MOIRAI$_{\text{Small}}$	**0.655**
w/o patch size constraints	0.720
w/o multi patch size	1.156
w/o Any-variate Attention	0.904
w/o mixture distribution	0.740
w/o LOTSA	0.809
w/o packing	0.785



(a) Mixture distribution.



(b) Student's t-distribution.

05 Conclusion

MOIRAI 作为一种基于掩码编码器的通用时间序列预测 Transformer，它一定程度上缓解了通用预测范式中面临的问题。通过利用大规模数据预训练的力量，这个时间序列基础模型彻底改变了以往每个数据集一个模型的格局。它为下游预测任务中的用户提供了巨大的优势，消除了使用深度学习模型实现准确预测所需的额外数据、大量计算资源的需求。

尽管 MOIRAI 在分布内和分布外性能上取得了显著成果，但这只是通用预测范式中的第一步。由于资源限制，研究者几乎没有进行超参数调整。

目前，研究者们开源了预训练数据，预训练代码以及模型权重为这一新兴研究领域的发展铺平了道路。